Controlled Substance Designation and Medical Substitution: Evidence from Gabapentin Monitoring and Upscheduling

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October 31, 2025

Disclaimers etc.

- This research was supported in part by the National Institutes of Health through the National Institute on Drug Abuse Justice Community Opioid Innovation Network cooperative through award number UG1DA050070 and the Leadership Education in Adolescent Health (LEAH) Program at Indiana University School of Medicine Division of Adolescent Medicine, through HRSA grant 5 T71MC45699-02-00.

- The opinions expressed in this article are the authors' own and do not reflect the view of the National Institutes of Health, the Department of Health and Human Services, or the United States government.

- All errors are our own

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➢ Led to being the 7th most prescribed drugs in the U.S.

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- New concerns around safety: Misuse potential and dangerous co-prescriptions with opioids

AP WORLD U.S. POLITICS SPORTS ENTERTAINMENT BUSINESS SCIENCE FACT CHECK

FDA warns of breathing risks with popular nerve drugs

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- New medical studies connecting use to dementia risk


MEDICINE

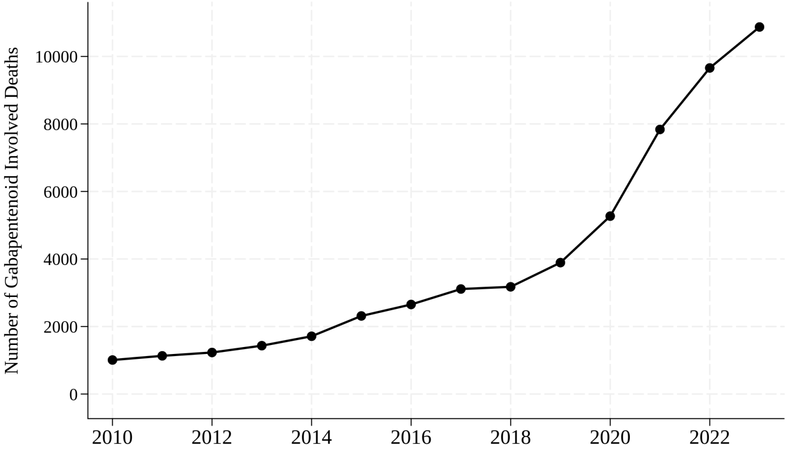
Commonly prescribed pain drug can increase your risk of dementia by up to 40%

By Tracy Swartz

Published July 11, 2025, 1:56 p.m. ET

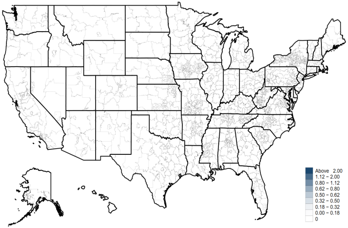
NEW YORK POST

Motivation - Gabapentanoid-Involved Mortality

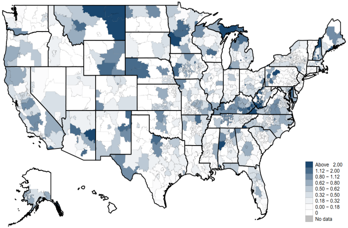


Motivation - Gabapentin-Involved Exposures

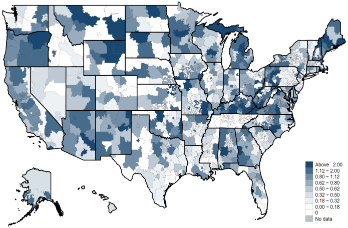
Poison Control Center Call Rates per 100,000



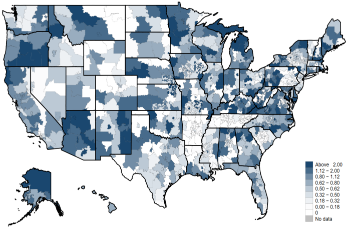
2011



2012



2014



2016

Motivation - State Policy Responses

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  - Requires refill limits, mandatory monitoring, etc.
  - Associated with $\downarrow$ in Medicare prescribing (Grauer & Cramer, 2022)
  - Similar upscheduling increased substitutions (Gupta et. al., 2023)

Research Question

- **Research Question:** How do policies restricting access to gabapentin impact:
  1. Gabapentin prescribing
  2. Opioid prescribing & co-prescriptions
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  4. Downstream overdose mortality

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- **Findings:**
  - Schedule V classification significant and lasting $\downarrow$ gabapentin prescriptions
    - No effect from DOC restriction
  - No evidence of substitution to higher opioid use
    - But, $\downarrow$ in co-prescriptions
  - Insignificant aggregate effects on other potential substitutions
    - Suggestive evidence of substitution among gabapentin-naive
  - $\downarrow$ Mortality involving both gabapentinoids and opioids

Contributions

- Policy signals that alter provider behavior
  - Changes in perceived risk or scrutiny
  - E.g. FDA drug withdrawals and black-box warnings, Consumer Production Safety Commission recalls
  - Parkinson et al., 2014; Collins et al., 2013; Freedman et al., 2012; Cawley and Rizzo, 2008
- Informational role of PDMPs
  - Reveal patient-specific histories to update provider beliefs
  - Bao et al., 2016; Buchmueller and Carey, 2018; Meara et al.,2016)

Contributions

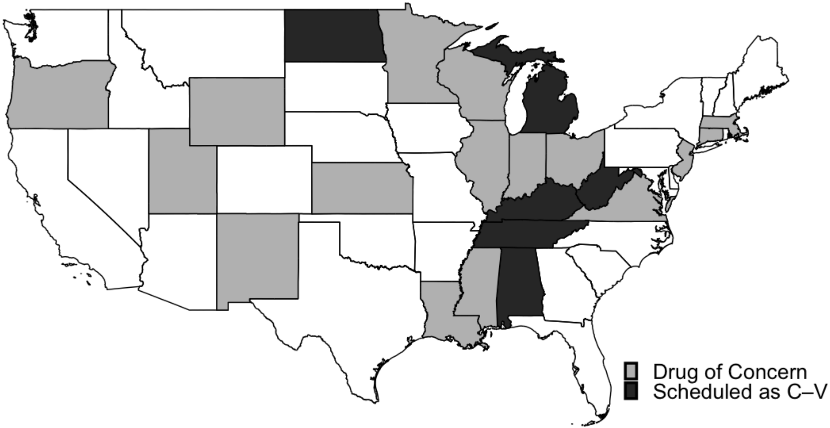
- Hassle costs in health policy
  - Administrative burdens may reduce utilization
  - E.g. Prior authorization
  - Finkelstein et al., 2019; Alpert et al., 2024; Sacks et al., 2021
- Consequences of supply side constraints on pain prescribing
  - Potential substitution to unscheduled or less regulated alternatives
  - May equalize hassle cost to more harmful substitutes
  - Gupta et al. (2023); Grauer and Cramer (2022)

Policy Background: Tiers of Prescribing Restrictions

<table>
<tr><th>Prescription drugs</th><th>Drug of Concern</th><th>Schedule V Controlled Substance
(Criterion: medical use, dependence and misuse potential)</th></tr>
<tr><td>1. Little/ no known risk of abuse. E.g. Antibiotics</td><td>1. Minimal abuse potential E.g. mild narcotic cough medicines</td><td>1. Low abuse potential E.g. Benzos, tramadol</td></tr>
<tr><td>2. Requires valid prescription</td><td>2. Prescription mostly valid for 12 mo.†</td><td>2. Prescription mostly valid for 12 mo.†</td></tr>
<tr><td colspan="3">3. Electronic/ phone scripts allowed w/o patient-provider in-person contact
4. Refill as authorized by practitioner
5. Delegate prescribing allowed</td></tr>
<tr><td colspan="2">6. Not require prescriber DEA number</td><td>6. Other general restrictions:
• Require prescriber DEA number
• Prescriptions must include patient name, patient DOB, drug name, dosage, strength, days of supply and refills (if any)
• Prescription bottle must carry FDA warning label prohibiting transfer
• Volume restrictions on wholesale distribution
• Locked storage in pharmacy
• Criminal penalties for unlawful or unlicensed distribution, possession, and use (without valid prescription)</td></tr>
<tr><td colspan="3">Increasing restrictions on prescribing</td></tr>
</table>

†With state ability to specify alternative validity limits. ‡For instance, Kentucky mandates that prescription be filled within 60 days.

Policy Background: Variation Across States



Timing of Policies

Conceptual Framework

➢ Extending Alpert et. al. (2024) hassle cost vs. information model

Utility of prescribing to patient i:

$$U_i = \alpha B_i - \lambda C(r_i) - h(s, \kappa),$$

where:

- B_i: expected clinical benefit to patient i
- $C(r_i)$: perceived cost of prescribing
  - Based on patient's observed and inferred risk r_i
  - λ: physician's weight of this risk
- $h(s, \kappa)$: hassle cost from administrative/regulatory burden
  - Depends on policy environment in state s and PDMP features κ

Conceptual Framework

Implicit threshold rule:

$$\tau(r_i; s, \kappa) = \frac{\lambda C(r_i) + h(s, \kappa)}{\alpha}.$$

- Prescription issued if $B_i \geq \tau(r_i; s, \kappa)$
- Restrictive policies impact choice through:
  - Increase weight of risk via salience (λ)
  - Additional information on patient's history/risk ($C(r_i)$)
  - Added hassle of administrative burdens ($h(s, \kappa)$)

Conceptual Framework

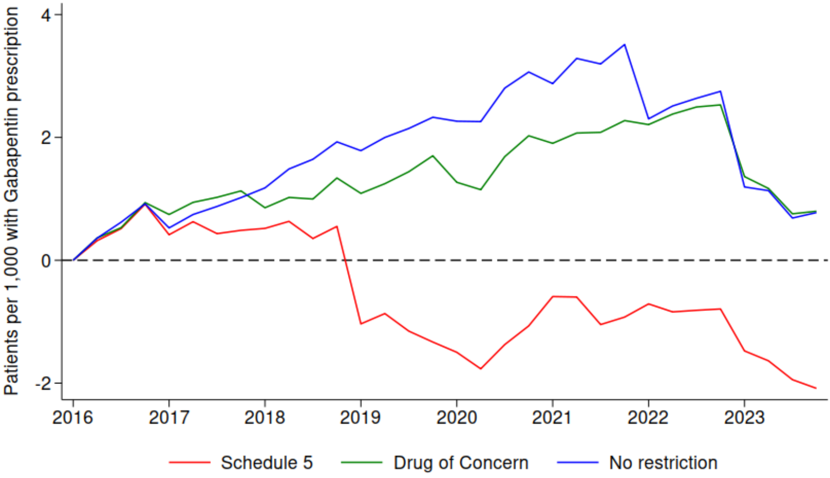
- Risk Salience
  - DOC without MA-PDMP
  - Likely stronger impact on naive vs. existing users
- Information Effect
  - DOC with MA-PDMP
  - Patient level transparency, including co-prescribing other controlled substance (risk)
- Hassle Cost
  - Schedule V classification
  - MA-PDMP, refill restrictions, increased visits, etc.
  - Fixed per-prescription cost:
    - Discourage extensive margin (fewer patients)
    - Encourage prescribing intensity ($\uparrow$ days' supply)

State-Quarter Descriptive Statistics

	N	Mean
Patients enrolled	1552	414,262.4
Gabapentin prescriptions per 1,000 patients	1552	12.47189
Pregabalyn prescriptions per 1,000 patients	1552	2.196201
Opioid prescriptions per 1,000 patients	1552	24.28274
Co-prescriptions of gabapentin and opioids per 1,000 patients	1552	3.025954
Anticonvulsant involved mortality per 100,000	2824	.3439374
Anticonvulsant and opioid involved mortality per 100,000	2824	.2765101
Gabapentin non-fatal poisoning diagnoses per 10,000 patients	1552	.388167
Non-pharmacological pain management per 1,000 patients	1552	56.87691

- Insurance Claims Data: Merative MarketScan® Research Databases 2016–2023 ($\sim$ 75 million unique enrollees)
- Mortality Data: National Center for Health Statistics' NVSS multiple cause-of-death file 2010-2023

Gabapentin Prescriptions by State Restriction Policy



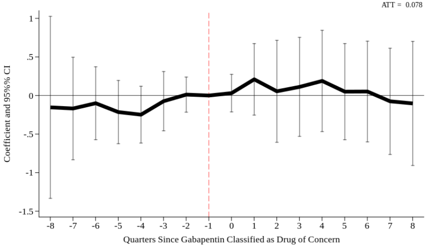
Empirical Model

$$Y_{g,s,t} = \alpha_s + \zeta_t + \eta_{g,t} + \sum_{\substack{k=-8 \\ k\neq -1}}^{8} \beta_k D^{(k)}_{g,s,t} + \varepsilon_{g,s,t},$$

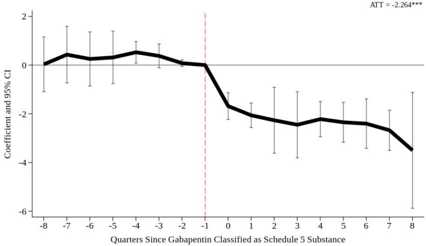
where:

- $Y_{g,s,t}$: gabapentin prescriptions per 1,000 population for state s in year-quarter t, when state is in implementation date cohort g
- $\alpha_s, \zeta_t, \eta_{g,t}$: state-by-cohort, year-quarter, cohort-by-quarter-year FEs
- $D^{(k)}_{g,s,t} = 1$ when state s is in event-quarter-year k relative to adoption quarter-year g
- Standard errors clustered at the state level
- Stacked difference-in-difference estimation for staggered treatment

Results - Gabapentin Prescribing



Drug of Concern

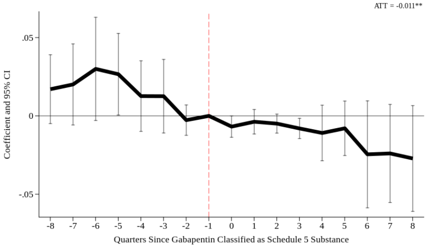


Schedule V

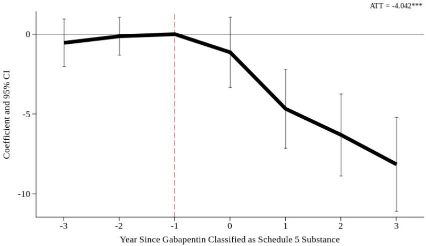
- Schedule V ATT $\approx$ 18% $\downarrow$ for a state with average prescribing rate

Robust to leaving out other policy states

Results - Gabapentin Prescribing - Public Payer

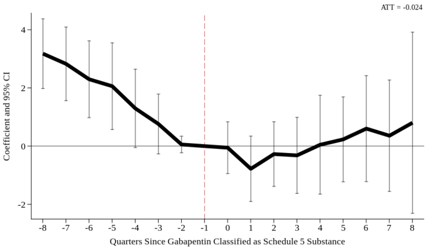


Medicaid SDUD Prescriptions

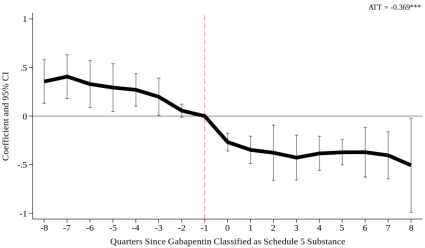


Medicare Part D Prescriptions

Results - Opioid Prescribing & Opioid/Gabapentin Co-Prescriptions



Opioid Prescribing

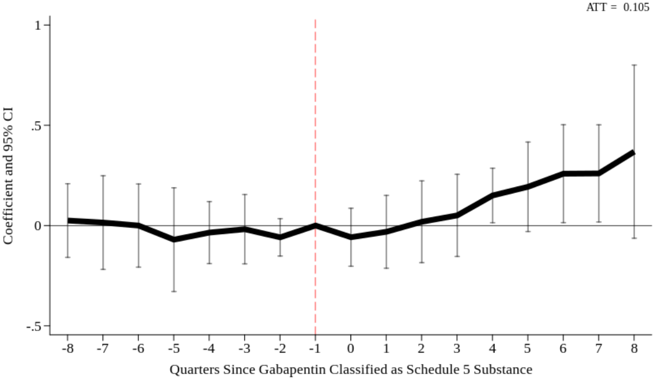


Co-prescriptions

- Schedule V ATT $\approx 12\% \downarrow$ for a state with average co-prescribing rate

No effects of DOC classification

Results - Substitutions

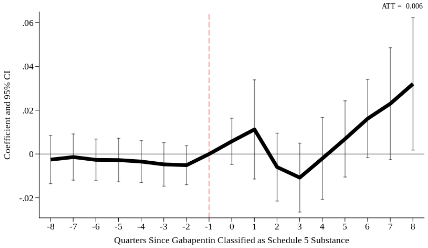


Pregabalin Prescribing

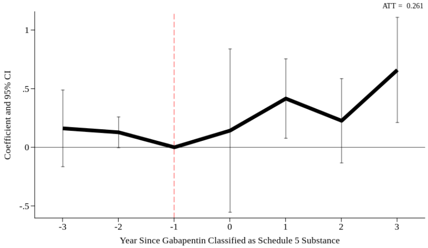
- Schedule V ATT $\approx$ 4.7% $\uparrow$ for a state with average Pregabalin prescribing rate

No effects of DOC classification | No effects on non-pharmacological pain management

Results - Substitution - Public Payer

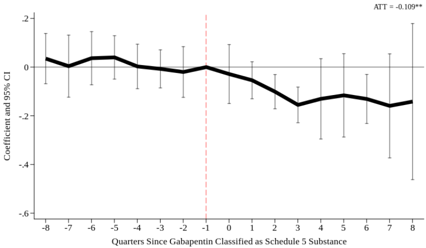


Medicaid SDUD Pregabalin

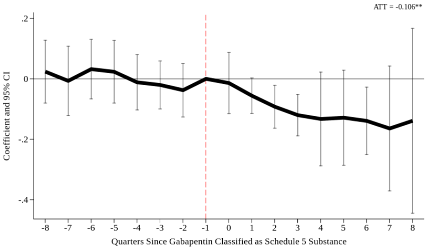


Medicare Part D Pregabalin

Results - Gabapentin Involved Mortalities



Gabapentinoids

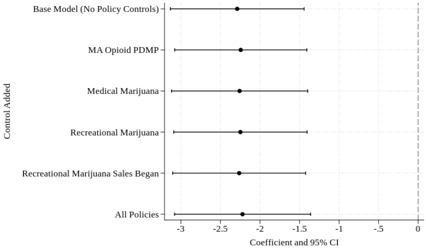


Gabapentinoids + Opioids

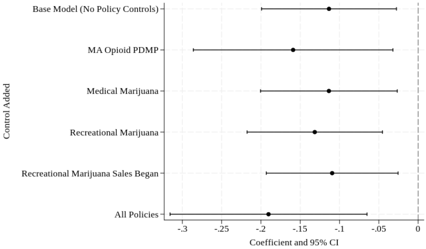
- Schedule V ATT $\approx 38\% \downarrow$ for a state with average gabapentinoid + opioid mortality rate

No effects of DOC classification | No effects on non-fatal adverse events

Robustness - Concurrent policies



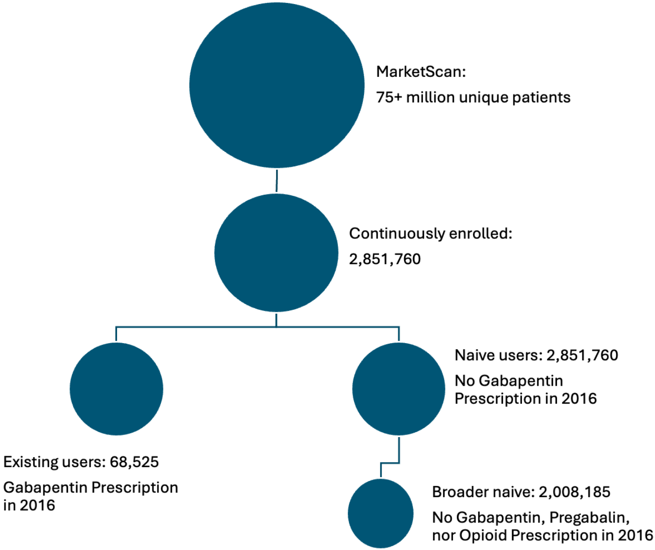
Gabapentin Prescribing



Gabapentinoids Mortality

Pregabalin and co-prescribing robust too

Cohort Analysis



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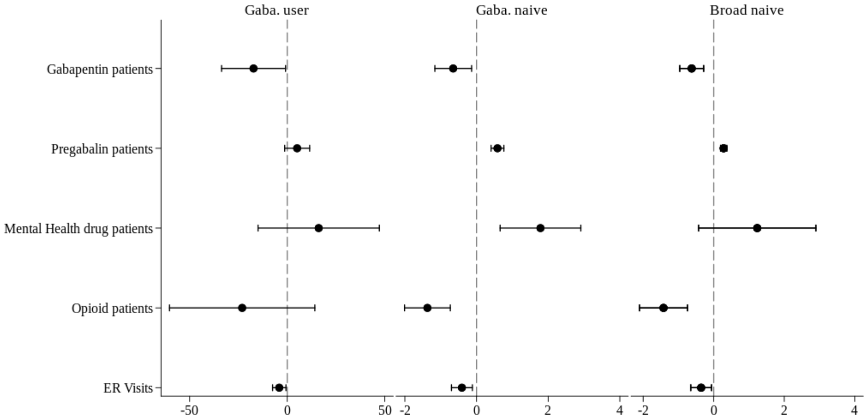

Empirical Tests of Mechanisms (in progress)

- Risk Salience: DOC without MA-PDMP
  - Prediction: decrease use, stronger impact on new users
- Information Effect: DOC with MA-PDMP
  - Prediction: more decreased use, stronger impact on "risky" users
- Hassle Cost: Schedule V
  - Predictions:
    - Largest decrease in overall use
    - Conditional on continued use, increase lengths of prescriptions

Empirical Tests of Mechanisms (in progress): Salience

- Salience Effect via Risk Perception Channel
  - No administrative regulations, but provided warning

For states **without** MA-PDMP:

$$Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta^{SAL} DOC_{st} + \varepsilon_{ist}$$

where:

- Y_{ist}: prescribing outcome for patient type i in state ss in quarter t
- $\beta^{SAL} < 0$: salience driven reduction in prescribing
- Stratify by existing-user and naive cohorts
  - Stronger for naive/new users = salience on initiation (vs continuation)

Empirical Tests of Mechanisms (in progress): Information

- Information Effect via Patient-Specific Signal Channel
  - Obtain patient-specific information with MA-PDMP

$$
\begin{aligned}
Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta_1^{INF} DOC_{st} \\
+ \beta_2^{INF} (DOC_{st} \times \textit{MA-PDMP}_s) \\
+ \beta_3^{INF} (DOC_{st} \times \textit{MA-PDMP}_s \times \textit{ExistingUsers}_i) + \varepsilon_{ist}
\end{aligned}
$$

where:

- $\textit{ExistingUsers}_i$: patients with prior gabapentin prescriptions
- $\beta_2^{INF} < 0$: DOC amplified in MA-PDMP state
- $\beta_3^{INF} < 0$: larger declines among high-risk patients
  - Consistent with providers incorporating new risk information

Empirical Tests of Mechanisms (in progress): Hassle

- Hassle-Cost Effect via Administrative Burden Channel
  - Obtain patient-specific information with MA-PDMP

$$\log(DaysSupply_{ist}) = \alpha_i + \delta_s + \theta_t + \beta^{HAS} CV_{st} + \varepsilon_{ist}]$$

where:

- $DaysSupply_{ist}$: length of prescription (intensive)
- Conditional on receiving the prescription
- $\beta^{HAS} > 0$: longer prescriptions to reduce admin. burden

Conclusion

- Examined prescribing consequences of restrictions on Gabapentin
  - Find significant and lasting declines with S-V classification, negligible impact of DOC classification
  - Limited evidence of clinically concerning substitution behavior
  - Decreases in potentially dangerous co-prescriptions and mortality rates
- Provides policy makers insight into:
  - Impacts of varying levels of strictness in restrictions (C-V vs. DOC)
  - Considerations for both drugs with misuse potential and common off-label usage

Thank you!

Questions?

Comments?

Suggestions?

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Policy Background: Implementation Dates

Scheduled as C-V		Drug of Concern	
State	**Date**	**State**	**Date**
Alabama	11/18/19	Connecticut	1/1/21
Kentucky	3/3/17	DC	6/7/19
Michigan	1/9/19	Illinois	7/7/23
North Dakota	4/10/19	Indiana	7/1/19
Tennessee	7/1/18	Kansas	5/11/18
Utah*	5/1/24	Louisiana	1/1/21
Virginia	7/1/19	Massachusetts	8/1/17
West Virginia	9/5/18	Minnesota	5/31/16
		Mississippi	5/7/21
		New Jersey	5/6/19
		New Mexico	2/28/23
		North Dakota	8/1/17
		Ohio	12/1/16
		Oregon	1/1/20
		Utah	12/9/21
		Virginia	2/23/17
		Wisconsin	9/1/21
		Wyoming	7/1/17

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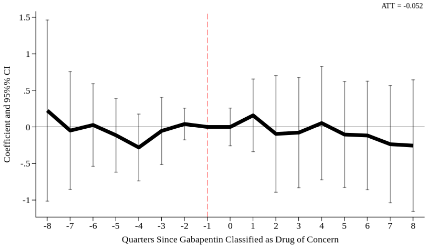
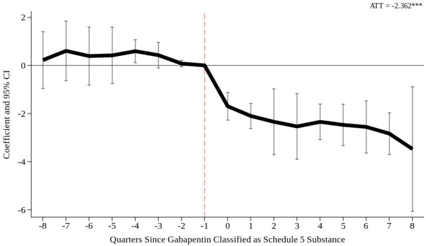
Results - Gabapentin Prescribing (drop other policy)



Drug of Concern



Schedule V

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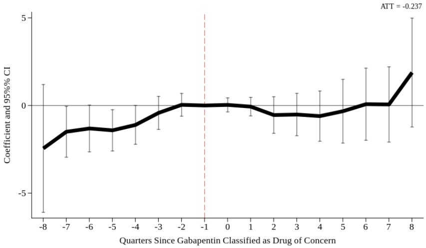
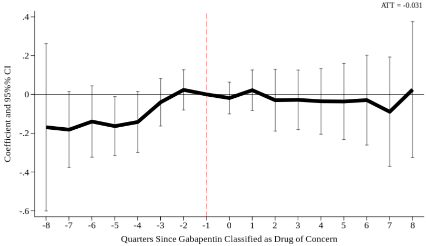
Results - Opioid Prescribing (DOC policy)



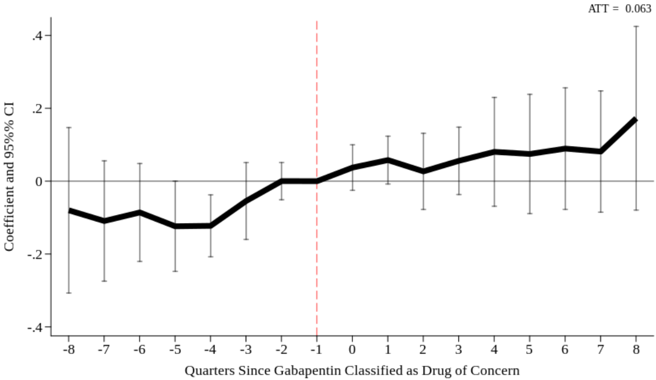
Opioid Prescribing



Gabapentin + Opioid Co-prescriptions

No effects of DOC classification

Results - Substitutions (DOC policy)



Pregabalin Prescribing

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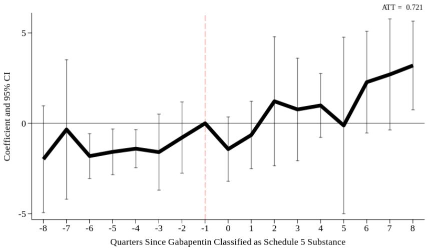
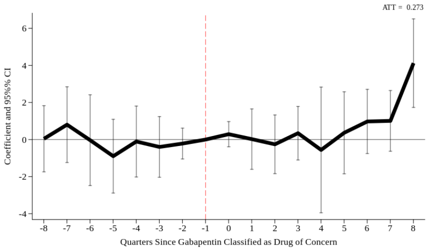
Results - Substitutions: Non-Pharmacological Pain Management



Schedule V



Drug of Concern

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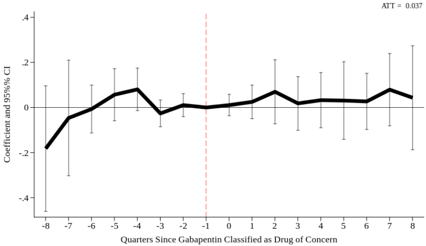
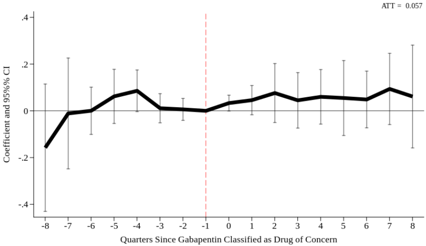
Results - Gabapentin Involved Mortalities (DOC policy)



Gabapentinoids



Gabapentinoids + Opioids

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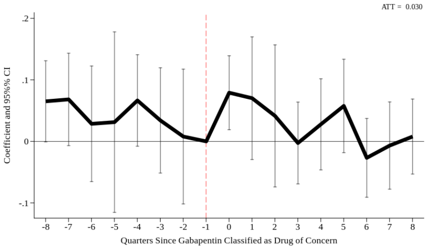
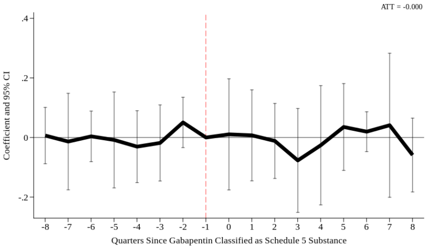
Results - Non-Fatal Gabapentin Involved Events



Drug of Concern



Schedule V

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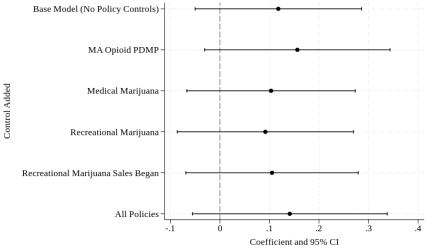
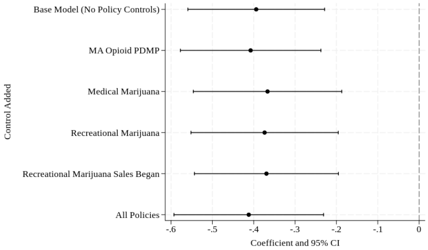
Robustness - Concurrent policies



Pregabalin Prescribing



Gabapentin + Opioid Co-Prescribing

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